# Cantera vs NN | Test IC 73 | NRMSE (util.py)

T0 = 1106.0 K  P0 = 1.911 atm  phi0 = 1.915  |  Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

| Model / same-state mean | Training | $T$ | $Y_{H_2}$ | $Y_H$ | $Y_O$ | $Y_{O_2}$ | $Y_{OH}$ | $Y_{H_2O}$ | $Y_{HO_2}$ | $Y_{H_2O_2}$ | $Y_{N_2}$ | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1 ANN10** | **10k** | 0.0381 | 0.0487 | 1.14 | 0.997 | 0.157 | 0.82 | 0.0187 | 0.373 | 6.14 | 0 | Complete |
| **2 P-only DeepONet10** | **10k** | 0.0376 | 0.025 | 0.757 | 0.734 | 0.154 | 0.77 | 0.019 | 0.37 | 0.648 | 0 | Complete |
| **2 P-only DeepONet10** | **long** | 3.35e-03 | 0.0147 | 0.0996 | 0.287 | 0.133 | 0.0551 | 0.0102 | 0.313 | 0.367 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | **10k** | 0.03 | 0.0143 | 0.592 | 0.58 | 7.95e-03 | 0.633 | 0.0124 | 0.0794 | 0.188 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | **long** | 4.06e-03 | 0.0152 | 0.103 | 0.297 | 0.14 | 0.0617 | 0.0108 | 0.321 | 0.367 | 0 | Complete |
| **4 T/P DeepONet10** | **10k** | 0.0466 | 0.0258 | 0.956 | 0.772 | 0.0161 | 0.986 | 0.0171 | 0.459 | 6.48 | 0 | Complete |
| **4 T/P DeepONet10** | **long** | 0.115 | 0.0565 | 2.29 | 2.96 | 0.0407 | 2.17 | 0.0443 | 0.14 | 0.326 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | **10k** | 0.0444 | 0.0276 | 0.93 | 0.673 | 0.0709 | 0.961 | 0.0159 | 0.422 | 2.4 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | **long** | 4.01e-03 | 6.44e-03 | 0.0809 | 0.135 | 0.0572 | 0.0809 | 4.74e-03 | 0.148 | 0.127 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | **10k** | 0.0497 | 0.0333 | 0.96 | 1.13 | 0.242 | 1.05 | 0.0279 | 0.634 | 0.879 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | **long** | 0.247 | 0.141 | 5.03 | 4.37 | 0.116 | 4.45 | 0.0868 | 8.52 | 1.37e+03 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | **10k** | 0.0248 | 0.0245 | 0.472 | 0.561 | 0.21 | 0.58 | 0.0195 | 0.528 | 0.672 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | **long** | 0.0744 | 0.0454 | 1.45 | 1.63 | 0.1 | 2.52 | 0.0462 | 0.284 | 28.5 | 0 | Complete |

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.